WRRV301
2026

Project Proposal

Crew name	**4SP_NET**	**How many crew members?**	☐ 3 ☐ 4
System Name	**Shuttle Management System**		
System Users	**Students, Marshals, Admin**		
Platform	☐ Desktop ☐ Mobile (Android only) ☐ Web system ☐ Hybrid (please specify i.e. website combined with a desktop or mobile interface) Click or tap here to enter proposed hybrid platform		

Briefly describe the system objective (project idea, problem to be solved)

The objective of the application is to address the problems currently faced by university students using shuttle services. Specifically try and find a solution to long waiting lines for shuttles which at times results in students being victims of crimes(their devices and other belongings getting taken). Allow for first years and other students that are not used to the shuttle environment and Gqeberha an easier way to find shuttle stops near their residences and an easier access to the schedule

The application will allow for students to select their stops, travel time available then book for a shuttle. Student will then be given details about the shuttle they will be taking such as the shuttle ID(could be a number plate) and a QR code. A student must be there ten minutes before departure and will be flagged if they don't show up for their booked shuttle, three flags resulting in suspension from using the shuttle services for a week. The QR allows for the taxi marshals to confirm that each student uses a shuttle they actually booked for. This also ensures that the shuttles are used by students that are actually from the university. The application will show stops(and stop info), travel routes and booking/travel times(schedules). The application will have a Find My Stop option that will guide students to stops near there residence. This is especially important for first year and other students not familiar with Gqeberha. Using the data collected from the first months, help with reassigning more shuttles to overloaded stops. The application will reassign shuttles dynamically, that is if one shuttle is fully booked assign the next 15 students to the next shuttle available at the stop.

Essentially, the application aims at doing away with waiting lines and maximising shuttle service efficiency .

List of entity classes (problem domain classes) – What data needs to be stored

Student, Shuttle, Shuttle Stop, Route, Schedule, Residence, Campus, Admin, Notification, Book

List of essential use cases – what is the basic functionality
Login / Logout NOT to be listed!!!

Register, View Stops, Book Shuttle, Book Shuttle, Available Shuttle,

Planned Implementation Tools (this could change once implementation starts)

Relational Database

☐ MS Access ☐ MySQL ☐ SQL Server ☐ SQLite

Development Environment (may select multiple)

☐ C# ☐ JAVA ☐ Android ☐ HTML ☐ CSS ☐ JavaScript

☐ Other (list other possible dev environments in brackets)

For official Use – DO NOT COMPLETE
Proposal Review

Proposal Outcome			25
Project Scope in line with Dev Crew Size ☐ Yes ☐ No	**Approved**	**Conditionally Approved**	**Rejected**

Criteria & Rating	Comments
Problem Overview Overview of the proposed system is detailed enough to make informed decision about feasibility of project (no reading between the lines) Lacks detail — Clear Overview 0 | 2 | 4 | 6 | 8 | 10	
Functionality & Reporting Sufficient number of create, report, update and delete (CRUD) use cases There must be evidence of some underlying computation/intelligence on data in addition to data retrieval (for reporting purposes specifically) Too few — Sufficient 0 | 1 | 2 | 3 | 4 | 5	
Data Support Sufficient amount of domain classes to map onto scope of project Lack of data — Sufficient data 0 | 1 | 2 | 3 | 4 | 5	
Planned Implementation Tools IDE should match the type of system to be implemented i.e. ASP for WEB No — Yes 0 | 1 | 2	
Innovation / Originality Novel idea - if a rehash of a previous system, then there needs to be enough added features Lacks originality — Innovative 0 | 1 | 2 | 3	